MANIPAL INSTITUTE OF TECHNOLOGY
BENGALURU
(A constituent unit of MAHE, Manipal)

B.Tech. Second Year
Course Plan: Theory Course

Department	Electronics & Communication Engineering	
Course Name & Code	Digital Circuits	ECE_2127
Semester & Branch	Third Semester, Jul-Dec 2025 Session	Electronics Engineering (VLSI Design & Technology)
Name of the Faculty	Dr. Balachandra Achar	
No of Contact Hours / Week	**L T P C** **3 0 0 3**	

Course Outcomes (CO) to PO, PSO and BL Mapping

COs	At the end of this course, the student should be able to	Contact Hours	Marks	Program Outcomes (PO)	Program Specific Outcomes (PSO)	Bloom's Level (BL)
CO1	Explain the basic concepts of number systems, Boolean algebra and various logical operations	3	8	1, 2	1	2, 3
CO2	Apply logic minimization techniques like K-map and Boolean algebra to design basic combinational circuits	5	14	1, 2, 3, 4	1	3
CO3	Construct combinational circuits including multiplexers, decoders and magnitude comparators using PLDs	11	31	1, 2, 3, 4	1	3
CO4	Analyse the operation of synchronous and asynchronous sequential circuits to understand their behaviour and performance	7	20	1, 2, 3, 4	1	4
CO5	Differentiate between various types of flips-flops and sequential circuit models to optimize the design and implementation of state machines	10	27	1, 2, 3, 4	1	4
	Total	**36**	**100**			

Course Articulation Matrix

COs / POs & PSOs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	1											3		
CO2	3	3	2	1									3		
CO3	3	3	2	1									3		
CO4	3	3	3	1									3		
CO5	3	3	2	1									3		
Average Articulation Level	**3**	**2.6**	**2.2**	**1**									**3**		

ICT Tools Used in Delivery and Assessment

Sr. No.	Name of the ICT tool used	Details of how it is used
1	Lighthouse (Brightspace) Learner Management System (LMS)	Assignment submission and grading tool. Delivery and assessment of continuous assignments on a weekly basis. Every student receives personalized values. Students must enter their answers, which will be auto-evaluated according to the provided expressions.
2	E-Pad: Paperless Digital Exams by LMS exams	Digital exam tool for the process of administering, taking, and grading mid-term and end-term examinations. Eliminates human errors in paper correction and offers detailed insights into student performance and exam results.
3	Microsoft Teams	Used as a live lecture tool, online doubt-clearing sessions, conducting tutorials as per the needs of students. Sharing the course contents & other resources.
4	Microsoft PowerPoint	Organized content delivery, formative assessment, supplemental handouts & notes, visualizing data with graphs and charts. Collaborating with colleagues in real-time, recording & sharing for review & self-study.
5	Microsoft SharePoint	Cloud storage and file sharing, sharing big video files and other resources among colleagues and learners.

Assessment Plan

In – Semester Assessments							
SN	**Assessment Method**	**Time Duration**	**Marks / Weightage**	**Pattern**	**Typology of Questions**	**Schedule**	**Topics Covered**
1	Mid Semester Exam	90 min	30 marks	**Objective type:** 10 MCQs for 10 marks **Descriptive type:** 4 or 6 questions for 20 marks	L2 L3 L4	12-19 Sep 2025	L1-L18
2	Surprise Quiz	30 min	6 marks	Problem solving	L3, L4		
3	LMS Quiz	30 min	4 marks	Objective type	L2, L3	20-28 Aug	
4	Assignment-1	30 min	6 marks	Poster presentation / Mini project	L3, L4	13-17 Oct	
5	Assignment-2	30 min	4 marks	Descriptive	L2, L3	21-28 Oct	
End – Semester Assessment							
1	End Semester Exam	180 min	50 marks	5 questions of 10 marks each. Each question can have 3 sub-questions	L2, L3, L4	17-28 Nov	L1-L36

Lesson Plan

L / T No.	Topics	Course Outcome Addressed
L0	Discussion on Vision, Mission, PEO, PO/PSO and Course Outline, Course Plan, Assessment Plan	--
L1	Review of Binary, Octal, Hexadecimal number systems and complement notations	CO1
L2	Review of digital logic, Boolean algebra theorems and logical gates	CO1
L3	Sum of products, Product of sums, Minterms and Maxterms expressions	CO1
L4	Logic minimization using 2, 3, 4 & 5-variable K-maps	CO2
L5	Minimization examples with & without don't care conditions to obtain minimal SOP and minimal POS expressions	CO2
L6	Half-adder, Full-adder, Half-subtractor, Full-subtractor	CO2
L7	Ripple-carry adder, Carry lookahead adder	CO2
L8	Cascade of CLA adders, BCD adder	CO2
L9	Magnitude Comparator, 1-bit, 2-bit and cascade connections	CO3
L10	Decoders, 2:4 and 3:8 decoders using logic gates	CO3
L11	4:16 decoder using 2:4 and 3:8 decoders	CO3
L12	2-input and 3-input logic design using decoders, multiple output functions	CO3
L13	Encoder, 8:3 priority encoder	CO3
L14	Multiplexer, 16:1 mux using 4:1 and 8:1 multiplexers	CO3
L15	Logic design using multiplexers	CO3
L16	Logic design examples using mux and decoders	CO3
L17	Programmable logic devices (PLDs), PROM	CO3
L18	Programmable logic arrays (PLAs), 3-input logic design using PLAs	CO3
L19	Programmable array logic (PAL), multiple output functions	CO3
L20	Latches, SR latches using NOR/NAND, switch debouncer	CO5
L21	Gated SR latch, D latch, propagation delay, setup time, hold time	CO5
L22	Master-slave SR flip-flop, D flip-flop, JK flip-flop, 0's & 1's catching	CO5

L23	Edge triggered D flip-flop, positive & negative edge triggering, flipflop conversions from one type to another.	CO5
L24	Shift registers, SISO, SIPO, PISO & PIPO, universal shift register	CO5
L25	Binary ripple counters, up/down, design examples	CO5
L26	Synchronous counters, mode-N counters, shift-register counters	CO5
L27	Design of synchronous counters using JK flip-flops	CO5
L28	Design of synchronous counters using D, T and SR flip-flops	CO5
L29	Self-correcting counters, design examples	CO5
L30	Structure and operation of clocked synchronous sequential networks	CO4
L31	Analysis of clocked synchronous sequential networks	CO4
L32	Serial binary adder as Mealy and Moore networks	CO4
L33	Sequence generator, Sequence detector with overlapping and non-overlapping	CO4
L34	Structure and operation of asynchronous sequential networks Analysis of asynchronous sequential networks	CO4
L35	Races in asynchronous sequential networks, State assignment and Race-free assignments	CO4
L36	Synthesis of asynchronous sequential networks	CO4

***Self-directed Learning:**
Hazards, Essential Hazards, Pulse mode sequential circuits, Design of Hazard free circuits.
Simulation of combinational and sequential circuits and their test benches using Verilog HDL

Reference Books

1. Donald Givone, "Digital Principles and Design", Tata McGraw Hill, 2002.
2. William Fletcher, "An Engineering approach to Digital Design", Prentice Hall of India, 2009.
3. Samir Palnitkar, "Verilog HDL: A Guide to Digital Design and Synthesis," Prentice Hall, 2003.
4. Raffiquzzaman, Chandra, Modern Computer Architecture, Galgotia Publications, 2008
5. William Stallings, Computer Organization and Architecture, Ninth edition, Pearson Education, 2013

Faculty Members Teaching the Course

Section	Faculty	Abbreviation
A		
B		

Submitted by: **Approved by:**

Program Outcomes

B.Tech. graduates will be able to:

PO 1: ***Engineering knowledge:*** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialisation for the solution of complex engineering problems.

PO 2: ***Problem analysis:*** Identify, formulate, research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3: ***Design/Development of Solutions:*** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.

PO 4: ***Conduct investigations of complex problems:*** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5: ***Modern tool usage:*** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO 6: ***The engineer and society:*** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: ***Environment and sustainability:*** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and the need for sustainable development.

PO 8: ***Ethics:*** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9: ***Individual and team work:*** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10: ***Communication:*** Communicate effectively on complex engineering activities with the engineering community and with the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO 11: ***Project management and finance:*** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO 12: ***Life-long learning:*** Recognise the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes

Electronics & Computer Engineering Graduates will be able to:

PSO1. Apply the principles of electronics & computer engineering across diverse domains including electronic design, software engineering, computer networks, embedded systems, operating systems, image processing, artificial intelligence, and associated areas.

PSO2. Analyse real-world challenges in electronics and computer domain using latest hardware and software tools to arrive at cost effective and appropriate solutions.

PSO3. Create and validate pioneering methodologies in electronics and computer engineering to address societal challenges.